



Spring 2024

“Robots for Good”: Ten Defining Questions (Paper Review)

Southern University of Science and Technology

ROBOTS AND SOCIETY

“Robots for good”: Ten defining questions

Selma Šabanović^{1*}, Vicky Charisi², Tony Belpaeme^{3,4}, Cindy L. Bethel⁵, Maja Mataric⁶, Robin Murphy⁷, Shelly Levy-Tzedek⁸

Ten questions to guide reflection and assessment of the “good” in robotics projects are suggested.

In July 2023, the United Nations’ AI for Good Global Summit displayed more than 50 robots to members of the public, United Nations (UN) staff and policy-makers, and the media. Along with invited speakers from academia, industry, and government, these robots were meant to demonstrate how emerging technologies can benefit society. As robotics researchers participating in this meeting, we were struck by tensions in how social good was defined in relation to robotic technology (1). In particular, we noted several humanoid robots presented as imminent artificial general intelligence solutions to the rising need for elder care, supporting creativity, companionship, service, and assistance. Robots that were being partially remote-controlled were described as autonomous; some developers shifted responsibility for answering questions about the ethical impacts of their projects to the robots themselves. The resulting discourse around these robots did not provide an honest and clear vision of what robots can really do or how they could, realistically and responsibly, lead toward societal good.

This led us to question what “good” means in the context of robotics. Some robotic applications might seem inherently good for society, such as disaster relief, educational, health care, and assistive robots. All sorts of robotics research and development can, however, be “laundered” and cast as good by being tenuously related to these applications. We need to be much more specific and explicit about how robotics will lead to measurable benefits and for whom before we call it good for society. We first need to operationally define what good is, what values we embed in robotic technology, and how we can be transparent in the choices that we make. We need to consider how we are communicating the values, choices, and expected good outcomes related to robots to different

audiences, taking into account the audiences’ technology literacy, lived experiences, and expectations. We acknowledge that when we mention good, we are speaking about several “goods”—to the user, to the public, to science, to economics, and to national security, among others—that might be complementary or in conflict with one another.

We present a (nonexhaustive) series of questions that every robotics-for-good project should return to repeatedly throughout its development and that the broader community (you too, Reviewer 2) can use to critically consider projects that claim that they are for the societal good. These are also questions that robotics projects should explicitly address in the communication of the work. We caution that these questions are meant to help robotics developers be more reflective of the strengths and limitations of their work and are not meant as a checklist for certifying good.

1) How did you come up with a definition for good in your project, and what specific societal goods will it lead to? Your project might be motivated by an evidence-based demand pull from the target community, by an “innovation push” from foundational scientific advances, or by other motives.

2) Who is this actually good for? Identify the explicit end users that your robot can assist, such as affluent older adults who can afford the technology, as well as others who stand to benefit from the project implicitly, such as developers or employers. Also explain in what specific ways the project is good for these populations. For example, the robot might aim to enable functional improvement of motor/social skills in users (2) or lead to improved work efficiency in an organization.

3) Who is this project not good for and in what ways? Workers may experience a change in their workload when a robot is introduced,

or bias toward women may be encouraged by the deployment of gendered technologies (3). The robot may also not be accessible to certain populations because of cost, maintenance needs, or other design features (4).

4) How do you measure whether the outcome of the robot is good? Be specific about whether you need to track behavioral, functional, psychological, and/or social changes in users and how you will measure these changes (for example, via clinical scales, self-report questionnaires, interviews, or other methods) (5).

5) Which stakeholders were involved in the design? State whether end users, their caregivers, clinicians, educators (6), managers, policy-makers, and/or other groups were involved in the design process, how they were involved, and what stages of design they were involved in, including developing initial design ideas, iterative user testing, and final evaluations.

6) What are the limitations of the robot—what can it not do? Be transparent about the robot’s sensing and reasoning capabilities, particularly those that might be misinterpreted by users (such as displaying emotions) (7). This will ensure that everyone has a clearer view of what the robot can achieve and how its functions need to be supported by others, including users, informal caregivers, and staff who handle and maintain the robot.

7) What are the potential negative consequences of the robot? Consider whether the robot’s use may result in perpetuating biases (8), socially isolating users, causing psychological, emotional, or social dependence, or hidden costs for maintenance, wireless infrastructure support, training, and outages (9).

8) What structures need to be put in place to guard against the potential negative consequences and by whom? It may be necessary to develop policy guidelines and regulatory frameworks that can be created by the UN, IEEE, national and local governments, or other institutions to ensure safe, inclusive, and equitable use of robots (10).

9) How have you protected the users against the potential harms of the robot?

Consider how to address concerns regarding legal liability or include fail-safes for the robot’s operation with redundancy.

10) How and to what degree do you make the capabilities, limitations, potential consequences, and biases of the robot transparent to specific audiences? The design and presentation of the robot may be constructed to guard against misperception. On the other hand, you may choose not to be fully transparent if you expect misperception to produce greater benefit to users; in that case, you should make clear why the degree of transparency you choose in your design is good for users and society more broadly.

We recognize that robots are not inherently good or bad, which is why these questions focus not on the technology alone but on the people who develop it, their values and goals, the ways and contexts in which robots will be used, and the societal norms and regulations that need to be in place to ensure good (or at least better) outcomes. We recognize that these can be difficult questions to answer, but it is important to start asking them regularly and openly before robots come into broader use. We want to inspire developers to engage in ethical reflection and consideration of good very specifically,

not just as a general idea that is difficult to assess. Developers should be ready to assume responsibility through answering these questions regarding the later uses of the robots that they design and their consequences. Equally important, societal stakeholders should be more regularly involved in discussions about robot development to provide their knowledgeable perspectives; regulatory bodies and international institutions like the UN should insist on in-depth consideration of these tensions and how we can address them in the design and application of future robots, alongside environmental consideration of technology development. Just because we can build new robots does not mean we should, if we are not clear on exactly how they will be good.

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 | **FOCUS** | ROBOTS AND SOCIETY



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- A Focus Paper published in **Science Robotics** on 231129
 - [[Website](#)] | [[Paper](#)] | [[Video](#)]
- Core idea in one sentence:
 - *Ten questions to guide reflection and assessment of the “good” in robotics projects are suggested.*

AI for Good Global Summit

Accelerating the United Nations
Sustainable Development Goals

Geneva, Switzerland, CICG
6-7 July 2023

SUMMIT PROGRAMME

EXPERT ML WORKSHOPS (4-5 July)

In July 2023, the United Nations' AI for Good Global Summit displayed more than 50 robots to members of the public, United Nations (UN) staff and policy-makers, and the media. Along with invited speakers from academia, industry, and government, these robots were meant to demonstrate how emerging technologies can benefit society.





KIME V3
THE BARTENDER'S HUMANOID
Macco Robotics



FridgeGuide Ai
ALL-IN-ONE APP TO TRACK YOUR
GROCERIES, CREATE NEW RECIPES,
AND REDUCE GLOBAL FOOD WASTE
FridgeGuide Ai, Inc.

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SingularityNET

Centre for Development of
Telematics (C-DOT)



Envirobot
BIO-INSPIRED AMPHIBIOUS SNAKE
ROBOT FOR POLLUTION DETECTION
IN LAKES
Ecole Polytechnique Fédérale
de Lausanne



MW75 Neuro
BRAIN-COMPUTER INTERFACE
HEADPHONES
Neurable



Embodied Quadrotors
VISION-CONTROLLED QUADROTORS
FOR DISASTER RESPONSE
University of Zurich



Atalante X
WORLD'S FIRST SELF-STABILIZING
WALKING EXOSKELETON
Wandercraft

In July 2023, the United Nations' AI for Good Global Summit displayed more than 50 robots to members of the public, United Nations (UN) staff and policy-makers, and the media. Along with invited speakers from academia, industry, and government, these robots were meant to demonstrate how emerging technologies can benefit society.

Summit Speakers



António Guterres
Secretary-General
United Nations

*Video address



Doreen Bogdan-Martin
Secretary-General
International
Telecommunication Union (ITU)



Tedros Adhanom
Ghebreyesus
Director-General
World Health Organization
(WHO)



Lila Ibrahim
Chief Operating Officer
Google DeepMind



As robotics researchers participating in this meeting, we were struck by tensions in how social good was defined in relation to robotic technology (1). In particular, we noted several humanoid robots presented as imminent artificial general intelligence solutions to the rising need for elder care, supporting creativity, companionship, service, and assistance. Robots that were being partially remote-controlled were described as autonomous; some developers shifted responsibility for answering questions about the ethical impacts of their projects to the robots themselves. The resulting discourse around these robots did not provide an honest and clear vision of what robots can really do or how they could, realistically and responsibly, lead toward societal good.

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Robots

● This article is more than 10 months old

Never underestimate a droid: robots gather at AI for Good summit in Geneva

Ai-da, Desdemona, Nadine and Geminoid join world's largest gathering of humanoids to promote AI as force for good



📷 Ai-da, a contemporary artist robot, cites Yoko Ono and Ai Weiwei as influences. Photograph: D Woldu/ITU



Hannah Devlin *Science correspondent*

Thu 6 Jul 2023 01:00 EDT

◀ Share

Grace is a nursing assistant, Ai-da a contemporary artist, Desdemona a purple-haired rock singer and Nadine is on hand for companionship and conversation.

They are all at the world's largest gathering of humanoid robots, which is under way at the [United Nations AI for Good global summit](#) in Geneva.

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The *First* Robot Press Conference

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The 17 Sustainable Development Goals?



17 Goals

EXAMPLE

Goal 1: End poverty in all its forms everywhere



169 Targets

EXAMPLE

Target 1.4: By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance

230 Indicators

EXAMPLE

Indicator 1.4.1 Proportion of population living in households with access to basic services

EXAMPLE

Indicator 1.4.2 Proportion of total adult population with secure tenure rights to land, with legally recognized documentation and who perceive their rights to land as secure, by sex and by type of tenure

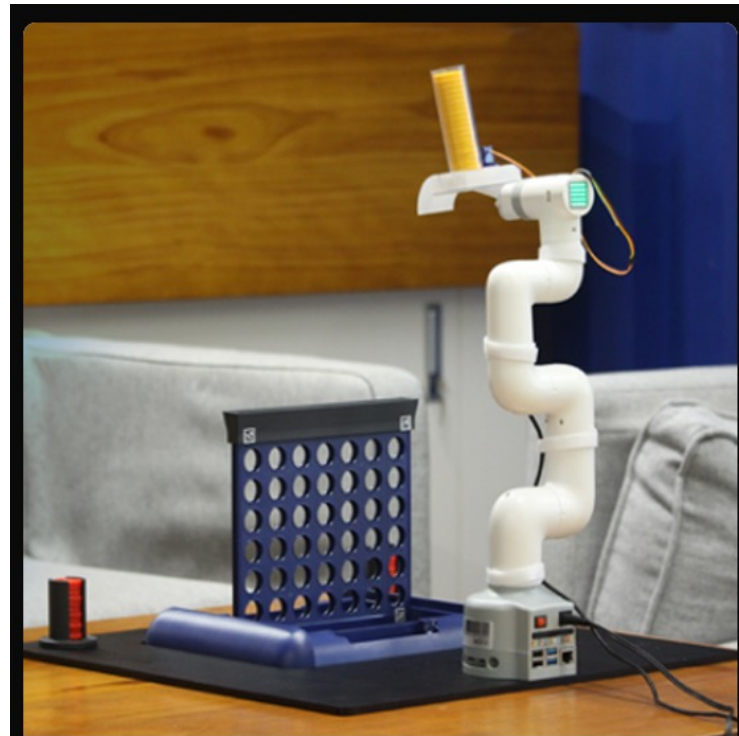
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We need to ...

- *Be more specific*
 - *Be more explicit*
 - *With measurable benefits*
 - *For whom*
-
- **Define ...**
 - The *good* design choices
 - **Communicate ...**
 - The (*robot*-related) design specifications
 - **Acknowledge ...**
 - The empathetic design trade-offs

We present a (nonexhaustive) series of questions that every robotics-for-good project should return to repeatedly throughout its development and that the broader community (you too, Reviewer 2) can use to critically consider projects that claim that they are for the societal good. These are also questions that robotics projects should explicitly address in the communication of the work. We caution that these questions are meant to help robotics developers be more reflective of the strengths and limitations of their work and are not meant as a checklist for certifying good.

1) How did you come up with a definition for good in your project, and what specific societal goods will it lead to? Your project might be motivated by an evidence-based demand pull from the target community, by an “innovation push” from foundational scientific advances, or by other motives.



myCobot 280

SIX-AXIS COLLABORATIVE ROBOT

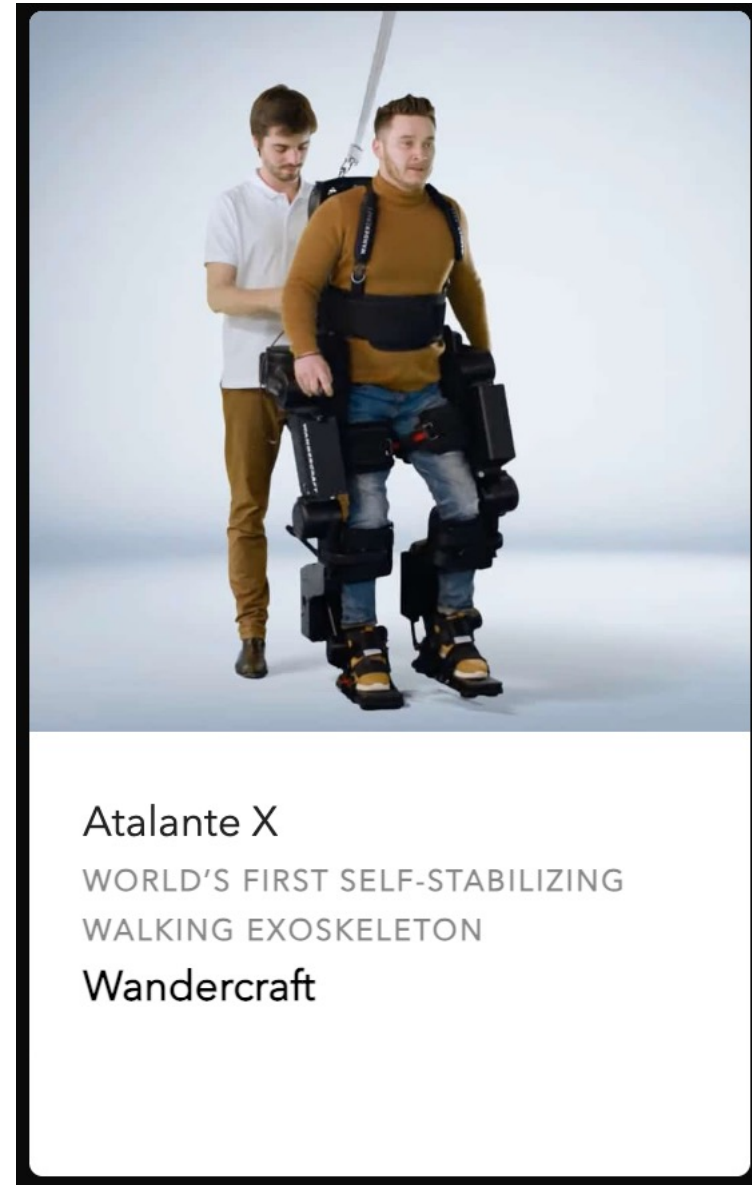
Elephant Robotics

1) How did you come up with a definition for good in your project, and what specific societal goods will it lead to? Your project might be motivated by an evidence-based demand pull from the target community, by an “innovation push” from foundational scientific advances, or by other motives.

- **Try it yourself:**

- *In this project, I define “good” as ...*
- *The motivation of my project is based on ...*
 - Evidence-based Demand Pull from a target community?
 - An “Innovation Push” from foundational scientific advances?
 - Other motives?

2) Who is this actually good for? Identify the explicit end users that your robot can assist, such as affluent older adults who can afford the technology, as well as others who stand to benefit from the project implicitly, such as developers or employers. Also explain in what specific ways the project is good for these populations. For example, the robot might aim to enable functional improvement of motor/social skills in users (2) or lead to improved work efficiency in an organization.



Atalante X

WORLD'S FIRST SELF-STABILIZING
WALKING EXOSKELETON

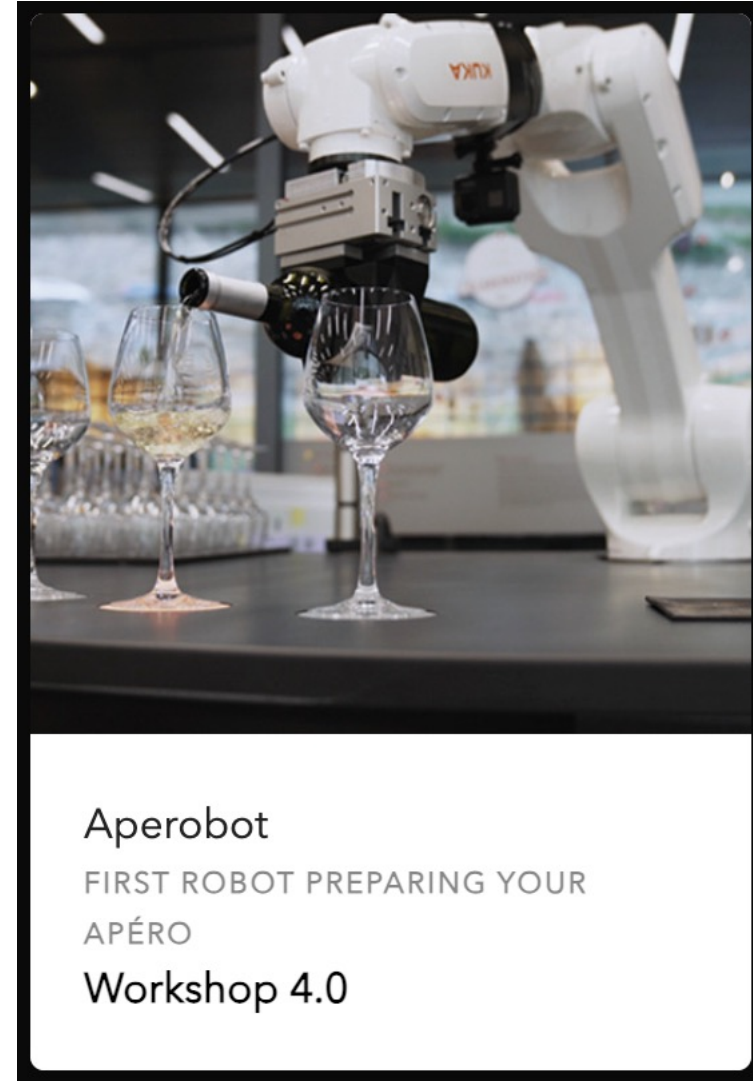
Wandercraft

2) Who is this actually good for? Identify the explicit end users that your robot can assist, such as affluent older adults who can afford the technology, as well as others who stand to benefit from the project implicitly, such as developers or employers. Also explain in what specific ways the project is good for these populations. For example, the robot might aim to enable functional improvement of motor/social skills in users (2) or lead to improved work efficiency in an organization.

- **Try it yourself:**

- *This project/robot is explicitly (~~actually~~) good for ...*
 - The explicit end user that your robot can assist
- *The project/robot can implicitly benefit ...*
 - Others who stand to benefit from the project implicitly
- *The project/robot is good for these populations in ways such as ...*
 - Examples with measurable benefits

3) Who is this project not good for and in what ways? Workers may experience a change in their workload when a robot is introduced, or bias toward women may be encouraged by the deployment of gendered technologies (3). The robot may also not be accessible to certain populations because of cost, maintenance needs, or other design features (4).

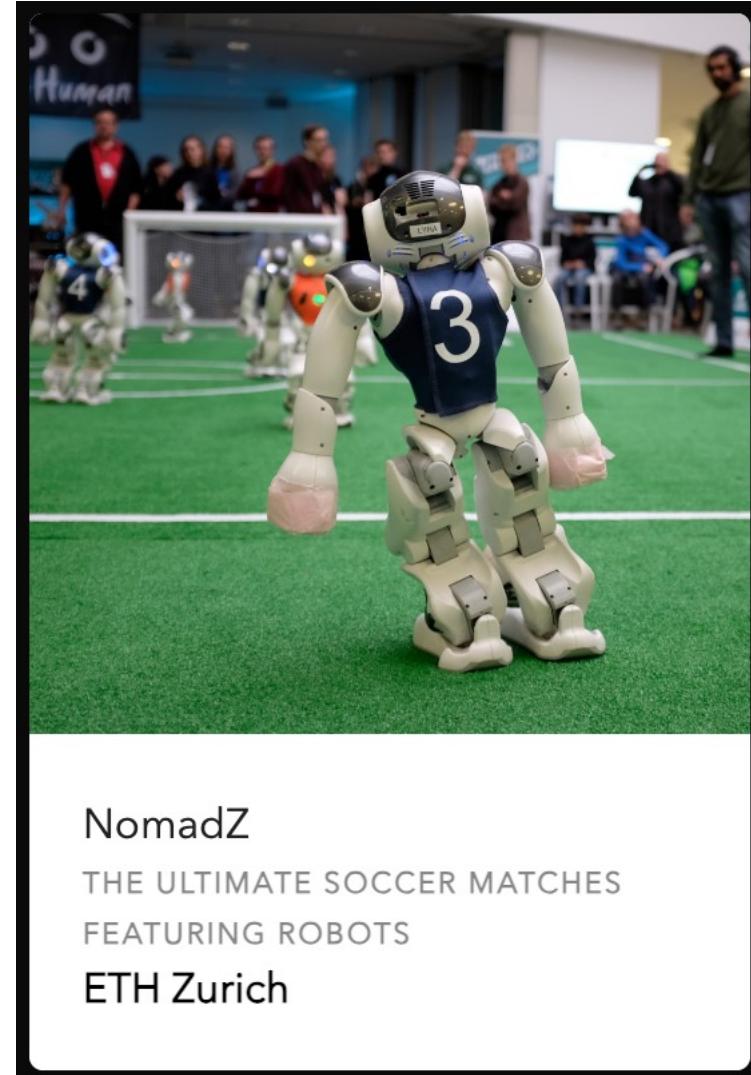


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- **Try it yourself:**

- *This project/robot is not good for ... in ways such as ...*
 - What changes could be anticipated?
 - Cost? Maintenance Needs? Other Design Features?

4) How do you measure whether the outcome of the robot is good? Be specific about whether you need to track behavioral, functional, psychological, and/or social changes in users and how you will measure these changes (for example, via clinical scales, self-report questionnaires, interviews, or other methods) (5).

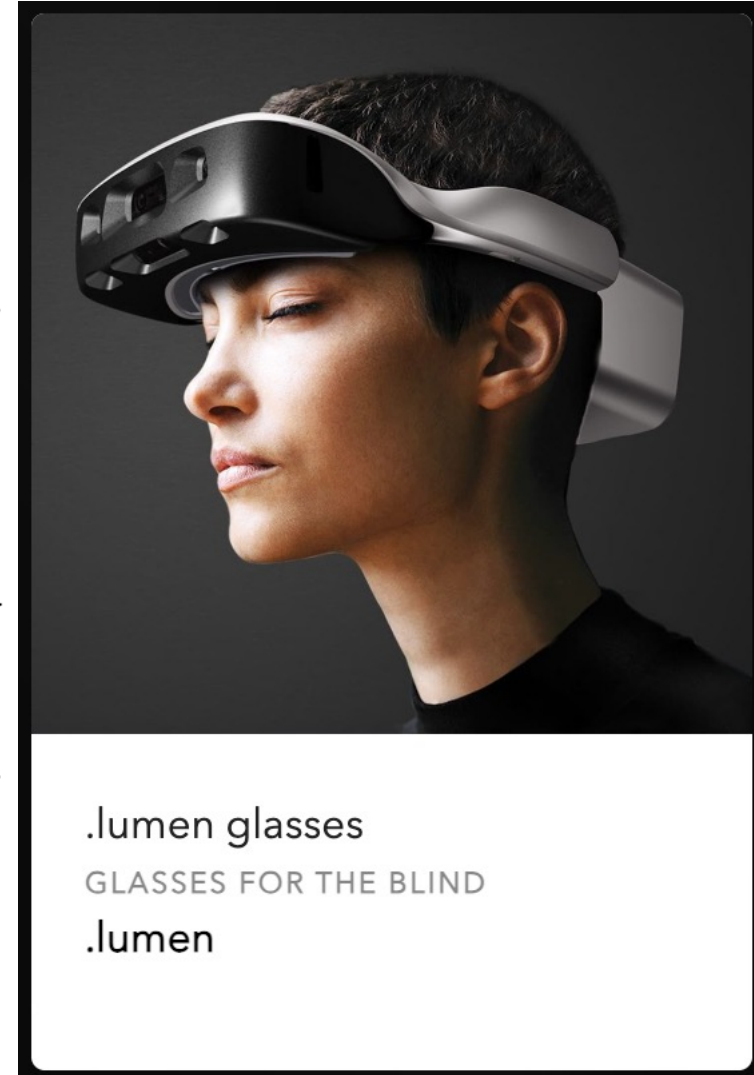


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- **Try it yourself:**

- *I measure the good outcome of the robot by ...*
 - Tracking the specific changes?
 - Behavioural / Functiona / Psychological / Social Metics?
 - In ways such as ...
 - Clinical Scales / Self-Report Questionnaires / Interviews / Other Methods?

5) Which stakeholders were involved in the design? State whether end users, their caregivers, clinicians, educators (6), managers, policy-makers, and/or other groups were involved in the design process, how they were involved, and what stages of design they were involved in, including developing initial design ideas, iterative user testing, and final evaluations.



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- **Try it yourself:**

- *The stakeholders involved in the design are ...*
 - The design process involves
 - The end users / their caregivers / clinicians / educators / managers / policy-makers / and/or other groups?
- *They were involved in ways such as ... during ... stage of design*
 - developing initial design ideas / iterative user testing / final evaluations?

6) What are the limitations of the robot—what can it not do? Be transparent about the robot’s sensing and reasoning capabilities, particularly those that might be misinterpreted by users (such as displaying emotions) (7). This will ensure that everyone has a clearer view of what the robot can achieve and how its functions need to be supported by others, including users, informal caregivers, and staff who handle and maintain the robot.



Aliengo

AUTONOMOUS EXPLORATION ROBOT

Unitree Robotics

6) What are the limitations of the robot—what can it not do? Be transparent about the robot’s sensing and reasoning capabilities, particularly those that might be misinterpreted by users (such as displaying emotions) (7). This will ensure that everyone has a clearer view of what the robot can achieve and how its functions need to be supported by others, including users, informal caregivers, and staff who handle and maintain the robot.

- **Try it yourself:**

- *The limitations of this robot are ... (what can it not do?)*
 - Be transparent about the robot’s sensing and reasoning capabilities, particularly those that might be misinterpreted by users (such as displaying emotions)
- *... thinks that the robot can ... and may require support from ...*
 - Does everyone (stakeholder) have a clear view of what the robot can achieve?
 - Does everyone (stakeholder) have a clear view of what support is needed?

7) What are the potential negative consequences of the robot? Consider whether the robot's use may result in perpetuating biases (8), socially isolating users; causing psychological, emotional, or social dependence; or hidden costs for maintenance, wireless infrastructure support, training, and outages (9).



ANYmal

WORLD'S FIRST EX-PROOF LEGGED
ROBOT

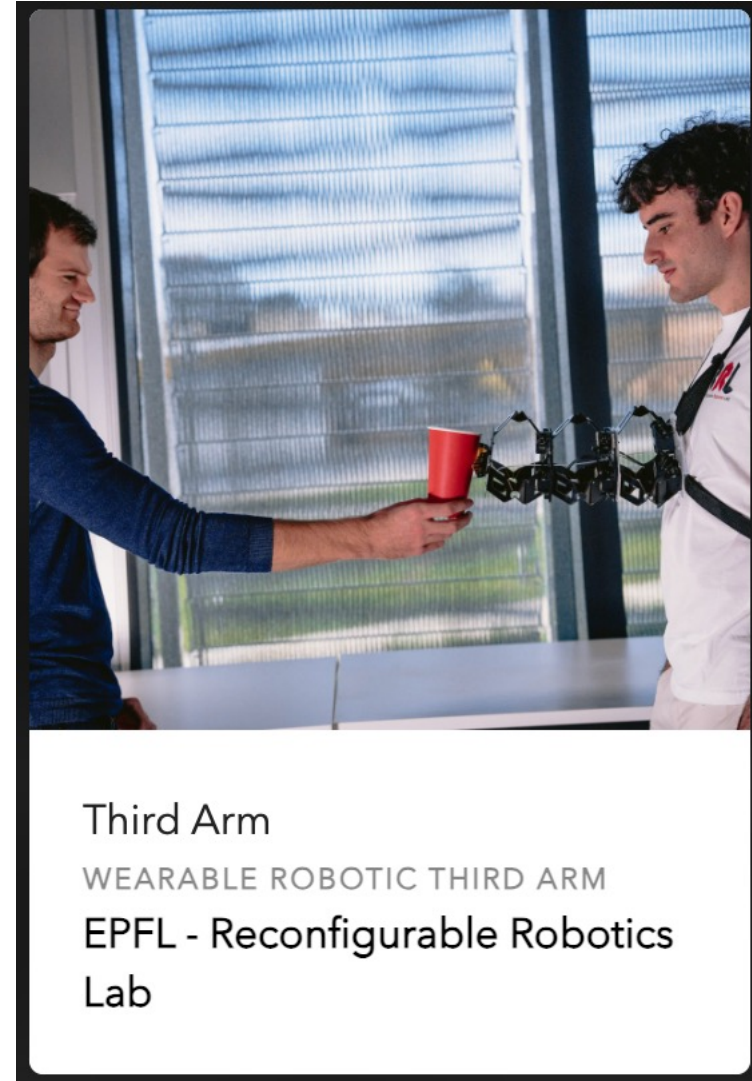
ANYbotics

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- **Try it yourself:**

- *The potential negative consequences of the robot are ...*
 - Consider whether the robot's use may result in
 - perpetuating biases, socially isolating users;
 - causing psychological, emotional, or social dependence; or
 - hidden costs for maintenance, wireless infrastructure support, training, and outages

8) What structures need to be put in place to guard against the potential negative consequences and by whom? It may be necessary to develop policy guidelines and regulatory frameworks that can be created by the UN, IEEE, national and local governments, or other institutions to ensure safe, inclusive, and equitable use of robots (10).



Third Arm

WEARABLE ROBOTIC THIRD ARM

EPFL - Reconfigurable Robotics
Lab

8) What structures need to be put in place to guard against the potential negative consequences and by whom? It may be necessary to develop policy guidelines and regulatory frameworks that can be created by the UN, IEEE, national and local governments, or other institutions to ensure safe, inclusive, and equitable use of robots (10).

- **Try it yourself:**

- *To guard against potential negative consequences by ..., we need to put in place structures such as ...*
 - It may be necessary to develop policy guidelines and regulatory frameworks that can be created by the UN, IEEE, national and local governments, or other institutions
 - to ensure safe, inclusive, and equitable use of robots

9) How have you protected the users against the potential harms of the robot? Consider how to address concerns regarding legal liability or include fail-safes for the robot's operation with redundancy.



PARO

THERAPEUTIC ROBOT BABY HARP
SEAL

National Institute of Advanced
Industrial Science and
Technology (AIST)

9) How have you protected the users against the potential harms of the robot? Consider how to address concerns regarding legal liability or include fail-safes for the robot's operation with redundancy.

- **Try it yourself:**

- ... *is used to protect the users against potential harms of the robot*
 - Consider how to address concerns regarding legal liability or include fail-safes for the robot's operation with redundancy.

10) How and to what degree do you make the capabilities, limitations, potential consequences, and biases of the robot transparent to specific audiences? The design and presentation of the robot may be constructed to guard against misperception. On the other hand, you may choose not to be fully transparent if you expect misperception to produce greater benefit to users; in that case, you should make clear why the degree of transparency you choose in your design is good for users and society more broadly.



Brain-Controlled Hand
Exoskeleton

BRAIN-CONTROLLED HAND
EXOSKELETON

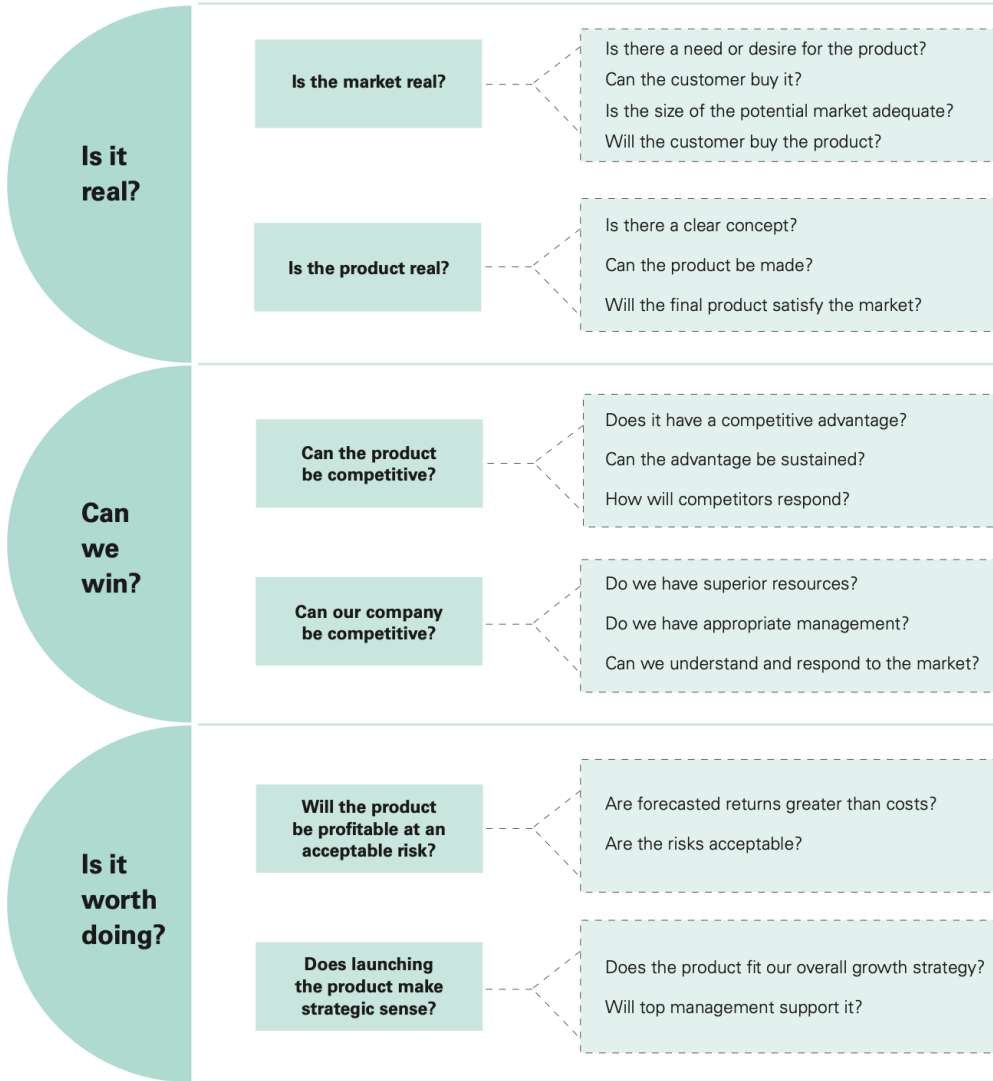
The University of Texas at Austin

10) How and to what degree do you make the capabilities, limitations, potential consequences, and biases of the robot transparent to specific audiences? The design and presentation of the robot may be constructed to guard against misperception. On the other hand, you may choose not to be fully transparent if you expect misperception to produce greater benefit to users; in that case, you should make clear why the degree of transparency you choose in your design is good for users and society more broadly.

- **Try it yourself:**

- *The capabilities / limitations / potential consequences / biases of the robot is transparent to ... audiences in terms of ... by ...*
 - How and to what degree?
 - How to manage misperception of the robot?

Is it Real? Can we Win? Is it Worth doing?



Real > Win > Worth Potential Questions and Scoring Continuums					
	Criterion/ Weight	0 (Uncertain?)	1	3	9
Real	Market Attractiveness Is the Market Real?	1.1. What is the need, want, or problem to solve? 1.2. Who are the identifiable customer(s) willing to buy? 1.3. How attractive is the market potential?	Unknown or no apparent need. No interest or no known fit with a customer or a market segment. Market Potential <\$4M	Not clear or not well defined need/want/problem to solve or market development required. Anecdotal customer(s) interest in buying the product/solution. Market Potential <\$8MM	Clearly defined and identifiable need/want/problem to solve. Well-defined customer(s) interest in buying the product/solution. Market Potential <\$20MM
	Technical Feasibility Is the Product Real?	2.1. What is the idea, concept, or solution that addresses the identified need? 2.2. To what extent do we have the technology and expertise to make it? (technical risk) 2.3. To what extent do we have the manufacturing or delivery capacity to provide it? (manufacturing risk) 2.4. How likely can we make the product within the defined market window? (timing risk) 2.5. To what extent will the product fit the customer's processes? (commercial risk)	Solution is not evident or no idea or concept currently exists that addresses the need. New Technology or invention is required. New Manufacturing Process or Capability needed. Not possible to meet market window. New customer manufacturing process(es) required.	Early ideas or concepts exist that address the need. Major development is required. (reformation, etc.) Major modification needed (equipment upgrades) Low probability in meeting market window. Major manufacturing process modification(s) required.	Several potential alternatives have been identified to solve the need. Minor development is required. (liner, thickness, color) Minor modification needed (process settings, etc.) Moderate probability of meeting market window. Minor manufacturing process modification(s) required.
	Product Advantage Is Product Competitive?	3.1. To what extent can our solution compete on design or performance features? 3.2. To what extent does the product complement or enhance an existing product offering? (product cannibalization risk) 3.3. To what extent does our price meet customer expectations? (pricing risk)	Has many attributes inferior to competition wrt key customer requirements. No expected enhancement to product, customer, or market position. Price has not been validated.	Is equal to but is sometimes inferior to competition wrt key customer requirements. Enhances product position at a specific customer only. Price has been validated by internal sources only (eg, sales rep).	Clearly exceeds competition wrt key customer requirements. Enhances specific market position and overall product portfolio. Price has been validated by customers representative of market segment.
	Synergies w. Core Competencies Is This Company Competitive?	4.1. To what extent does it leverage our core technology or build on an existing platform? (investment risk) 4.2. To what extent do we have a path to market and business model to be successful? (market share risk) 4.3. To what extent do we have the experience, skills, and human resources to be successful? (project risk)	Leverages none of core technology platforms and technical strengths. No Path to Market/New Market/No business model. No existing FTE resources available for project.	Leverages an insignificant level of core technology platforms and technical strengths. Elements missing from path to market/business model new customer(s) within existing market. Major FTE resource additions or skill set improvements are required.	Leverages some level of core technology platforms and technical strengths. Existing path to market, existing business model, expanded offering to current customer(s). Minor FTE resource additions or skill set improvements are required.
Worth	Strategic Fit Is it Strategic?	5.1. How well is this opportunity aligned with the strategic plan for our organization? 5.2. To what extent does this opportunity open the door to new business in the future? (strategic leverage) 5.3. To what extent are there overriding factors? (eg, affordability, can't afford NOT to do it?)	Business opportunity does not align with existing strategic objectives. No other business opportunities expected outside of this specific offering. No overriding factors.	Business opportunity is partially aligned with strategic objectives. Opportunity may be leveraged in more than one customer application. Small overriding factors.	Business opportunity is closely aligned with strategic objectives. Opportunity may be leveraged in several customer applications. Strong overriding factors.
	Financial Risk Is it Profitable?	6.1. What are the capital requirements? (financial risk) 6.2. What are the full-time equivalent requirements? (human capital risk) 6.3. What is the range for projected annual sales in year five? 6.4. What is the projected range for rate of return on capital invested? 6.5. What is the range for projected size of return in ten years? (net present value or NPV) 6.6. To what extent do we have confidence in the opportunity? (forecast risk)	Very Large (> \$370K) Very Large (>\$390K per year) <\$1MM (First Full Year) GM < 40% (Ten Year) NPV < \$0 Largely not validated (Confidence<30%)	Large (\$37K - \$370K) Large (\$260K - \$390K per year) \$1-2MM (First Full Year) GM = 40% - 55% (Ten Year) NPV = \$1K to \$800M Validated (Confidence>30%)	Moderate (\$1.5K - 37K) Moderate (\$65K - 260K per year) \$2-4MM (First Full Year) GM = 55% - 70% (Ten Year) NPV = \$801K to 3,900K Validated (Confidence>60%)
	Risk/Reward Is it Profitable?				
	Risk/Reward Is it Profitable?				

RWW Framework

- Is it **Real**? $\Rightarrow$ Q1/2/3
 - Is the the robot socially good?
 - Is the robotic technology real?
- Can we **Win**? $\Rightarrow$ Q4/5/6/7
 - Can the robot be competitive?
 - Can the stakeholder benefit?
- Is it **Worth** doing? $\Rightarrow$ Q8/9/10
 - Will the robot be socially good at an acceptable risk?
 - Does launching the robot make social sense?

*Can be further refined towards
Robot Design for Social Good*

1) How did you come up with a definition for good in your project, and what specific societal goods will it lead to? Your project might be motivated by an evidence-based demand pull from the target community, by an “innovation push” from foundational scientific advances, or by other motives.

2) Who is this actually good for? Identify the explicit end users that your robot can assist, such as affluent older adults who can afford the technology, as well as others who stand to benefit from the project implicitly, such as developers or employers. Also explain in what specific ways the project is good for these populations. For example, the robot might aim to enable functional improvement of motor/social skills in users (2) or lead to improved work efficiency in an organization.

3) Who is this project not good for and in what ways? Workers may experience a change in their workload when a robot is introduced, or bias toward women may be encouraged by the deployment of gendered technologies (3). The robot may also not be accessible to certain populations because of cost, maintenance needs, or other design features (4).

4) How do you measure whether the outcome of the robot is good? Be specific about whether you need to track behavioral, functional, psychological, and/or social changes in users and how you will measure these changes (for example, via clinical scales, self-report questionnaires, interviews, or other methods) (5).

5) Which stakeholders were involved in the design? State whether end users, their caregivers, clinicians, educators (6), managers, policy-makers, and/or other groups were involved in the design process, how they were involved, and what stages of design they were involved in, including developing initial design ideas, iterative user testing, and final evaluations.

6) What are the limitations of the robot—what can it not do? Be transparent about the robot’s sensing and reasoning capabilities, particularly those that might be misinterpreted by users (such as displaying emotions) (7). This will ensure that everyone has a clearer view of what the robot can achieve and how its functions need to be supported by others, including users, informal caregivers, and staff who handle and maintain the robot.

7) What are the potential negative consequences of the robot? Consider whether the robot’s use may result in perpetuating biases (8), socially isolating users; causing psychological, emotional, or social dependence; or hidden costs for maintenance, wireless infrastructure support, training, and outages (9).

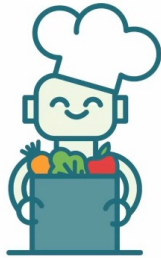
8) What structures need to be put in place to guard against the potential negative consequences and by whom? It may be necessary to develop policy guidelines and regulatory frameworks that can be created by the UN, IEEE, national and local governments, or other institutions to ensure safe, inclusive, and equitable use of robots (10).

9) How have you protected the users against the potential harms of the robot? Consider how to address concerns regarding legal liability or include fail-safes for the robot’s operation with redundancy.

10) How and to what degree do you make the capabilities, limitations, potential consequences, and biases of the robot transparent to specific audiences? The design and presentation of the robot may be constructed to guard against misperception. On the other hand, you may choose not to be fully transparent if you expect misperception to produce greater benefit to users; in that case, you should make clear why the degree of transparency you choose in your design is good for users and society more broadly.



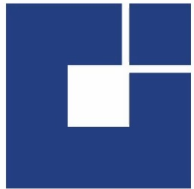
KIME V3
THE BARTENDER'S HUMANOID
Macco Robotics



FridgeGuide Ai
ALL-IN-ONE APP TO TRACK YOUR
GROCERIES, CREATE NEW RECIPES,
AND REDUCE GLOBAL FOOD WASTE
FridgeGuide Ai, Inc.



Desdemona
ROCK STAR ROBOT OF THE JAM
GALAXY BAND
SingularityNET



सी-डॉट
C-DOT

C-DOT
A PREMIER TELECOM R&D CENTRE OF
GOVERNMENT OF INDIA
Centre for Development of
Telematics (C-DOT)



Envirobot
BIO-INSPIRED AMPHIBIOUS SNAKE
ROBOT FOR POLLUTION DETECTION
IN LAKES
Ecole Polytechnique Fédérale
de Lausanne



MW75 Neuro
BRAIN-COMPUTER INTERFACE
HEADPHONES
Neurable



Embodied Quadrotors
VISION-CONTROLLED QUADROTORS
FOR DISASTER RESPONSE
University of Zurich



Atalante X
WORLD'S FIRST SELF-STABILIZING
WALKING EXOSKELETON
Wandercraft

Concluding Remarks

- **We recognize that robots are not inherently good or bad**, which is why these questions focus not on the technology alone but on the people who develop it, their values and goals, the ways and contexts in which robots will be used, and the societal norms and regulations that need to be in place to ensure good (or at least better) outcomes.
- **We recognize that these can be difficult questions to answer**, but it is important to start asking them regularly and openly before robots come into broader use.

Concluding Remarks

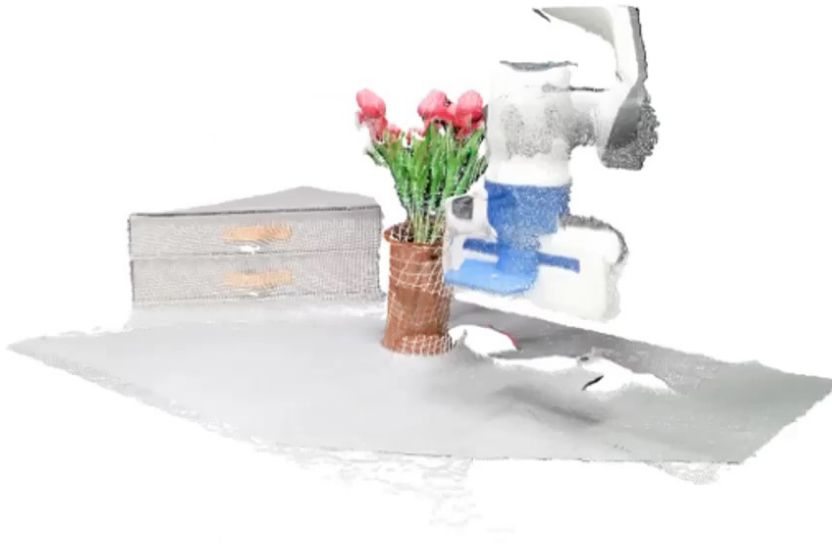
- **We want to inspire** developers to engage in ethical reflection and consideration of good very specifically, not just as a general idea that is difficult to assess. Developers should be ready to assume responsibility through answering these questions regarding the later uses of the robots that they design and their consequences.
- **Equally important**, societal stakeholders should be more regularly involved in discussions about robot development to provide their knowledgeable perspectives; regulatory bodies and international institutions like the UN should insist on in-depth consideration of these tensions and how we can address them in the design and application of future robots, alongside environmental consideration of technology development.

Just because we can build
new robots *does not mean*
we should, if we are not
clear on exactly how they
will be good.

Foundation Models for Robotic Manipulation

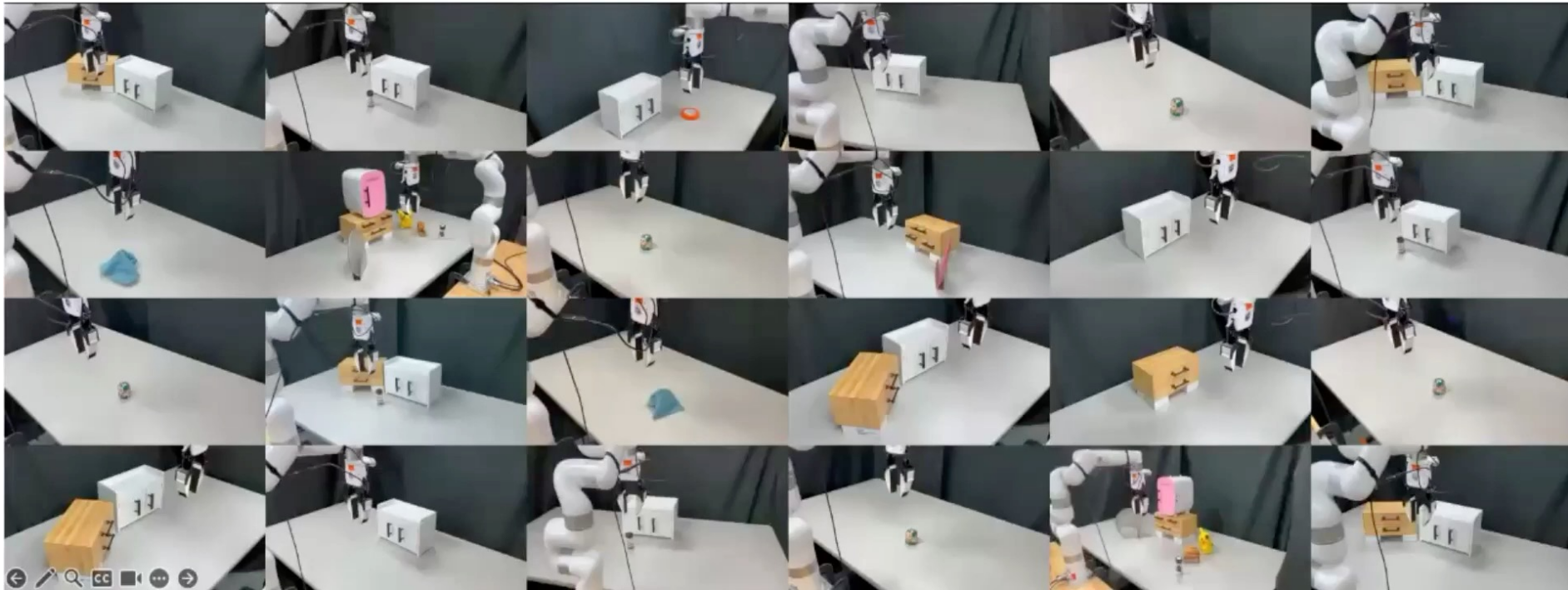


Open the top drawer.



Foundation Models for Robotic Manipulation

RoboEXP: Action-Conditioned Scene Graph via Interactive Exploration for Robotic Manipulation





Spring 2024

Thank you~

Southern University of Science and Technology